MIL-P-51474A(EA)
25 April 1984
SUPERSEDING
MIL-P-51474(EA)
16 June 1980

MILITARY SPECIFICATION

PROJECTILE, 64 MM, RIOT CONTROL, KE M743

This specification is approved for use by US Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and packaging for M743 projectile, selected components, and the M743, 64 MM projectile assembled together with the M755 blank cartridge in a carrier assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-1168 - Lot Numbering of Ammunition.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWING AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

TDPL 122-3-60 - Projectile, 64 MM: Riot Control, KE M743 w/ctg M755 (6 each).
122-3-70 - Packing and Marking Diagram-Projectile 64 MM and M743 w/ctg M755.
122-3-43 - Cartridge 5.56 MM: Blank M755.
122-3-75 - Projectile, 64 MM - M743.

US ARMY MATERIEL DEVELOPMENT AND READINESS COMMAND

DARCOM Drawing 19-48-4116-2OPA1OO2 - Unitization Procedure for Boxed Ammunition and Components on 4-Way Entry Pallets.
DARCOM Drawing 19-48-4116/32-20PA1002 - Appendix 32, Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets.

TECHNICAL MANUAL

TM9-1010-224-10 - Operator's Manual, Launcher, Projectile, 64 MM Riot Control.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D 3951 - Commercial Packaging, Practice for DoD Adopted.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 122-3-60 and on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the projectile shall conform to the specifications and drawings listed on TDPL 1.22-3-60.

3.2 Assembly. The projectile shall be assembled as shown on Drawing 122-3-75.

3.3 Performance.

3.3.1 Wrap assembly. The wrap assembly shall have a maximum yield force of 24.25 pounds and a maximum yield strain of .039 inch per inch (in/in) when tested in accordance with 4.4.4.1.

3.3.2 Integrity/accuracy. When fired from a launcher, the projectile shall have no separation of the breakband from the projectile prior to target impact and shall maintain sufficient accuracy to strike a 46 centimeter (cm) wide, 175 cm high target at a distance of 40.0 + 0.5 meter from the launcher when tested in accordance with 4.4.4.2.

3.5 Preproduction. Prior to the start of regular production, a preproduction sample of components shall be produced in accordance with this specification for examination and test (see 4.3).

3.6 Workmanship. The projectile body shall be free from imperfections listed in Table I of MIL-STD-407 and damage such as rips or tears which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified

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in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Preproduction lot. A preproduction lot of projectiles and components shall be manufactured using the same methods, materials, equipment and processes as will be used in regular production. The preproduction lot shall consist of the quantity of the projectiles and components and conform to corresponding drawings listed as follows:

Quantity	Item	Drawing
100	Airfoil bodies (a minimum of ten bodies from each mold cavity)	122-3-74
200	Projectiles	122-3-75
48	Projectiles with cartridges	122-3-60
48	projectiles, packed level B	122-3-70-10

4.3,2 Sampling.

4.3.2.1 For examination. The preproduction samples of projectiles and components listed below shall be randomly taken from the preproduction lot.

Quantity	Item	Drawing
10	Airfoil bodies (a minimum of one from each mold cavity)	122-3-74
10	Projectiles	122-3-75
24	Projectiles with cartridges	122-3-60
24	Projectiles, packed level B	122-3-70-10

4.3.2.2 For destructive tests.

(a) Wrap assembly. A sample of 20 completely assembled projectiles shall be randomly taken from the preproduction lot.

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(b) Integrity and accuracy. A sample of 32 completely assembled projectiles shall be randomly taken from the preproduction lot.

4.3.3 Inspection procedure.

4.3.3.1 For examination. The sample from the preproduction lot shall be examined for all the requirements specified on the drawings and specification The acceptance number is zero.

4.3.3.2 For destructive tests.

(a) Wrap assembly. The sample shall be tested in accordance with MIL-STD-414 and 4.4.4.1 using and AQL of 1.0 percent defective.

(b) Integrity/accuracy. The sample shall be tested in accordance with 4.4.4.2. The preproduction lot shall be accepted on 2 or less defective in the sample or rejected on 3 defective in the sample.

4.3.4 Acceptance/rejection criteria. The preproduction lot shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Qaulity conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. A lot shall consist of no more than 5016 projectiles produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions. No more than one lot number of cartridges shall be packaged with any one lot of projectiles.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive test.

4.4.2.2.1 Wrap assembly. Sampling shall be conducted in accordance with MIL-STD-414, level II.

4.4.2.2.2 Integrity/accuracy. Sampling shall be conducted in accordance with MIL-STD-105, level S4.

4.4.3 Inspection procedure.

4.4.3.1 For lamination. Sample projectiles and level B packaging (section 5) shall be examined in accordance with the classification of defects and with MIL-STD-105 .

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4.4.3.2 For destructive test.

4.4.3.2.1 Wrap assembly. Wrap assemblies shall be tested in accordance with 4.4.4.1 and evaluated with MIL-STD-414 at an AQL of 2.5 percent defective.

4.4,3.2.2 Integrity/accuracy. Projectiles shall be tested in accordance with 4,4.4.2 and evaluated with MIL-STD-105 at AQL at 6.5 percent defective.

4.4.3.3 Classification of defects.

(a) Projectile, 64 MM- M743 (Dwg 122-3-75).

Categories	Defects
Critical:	None defined
Major	AQL 1.0 percent defective
101	Assembled improperly
102	Outside diameter incorrect
103	Inside diameter incorrect
104	Weight incorrect
105	Tissue wrap incorrect

(b) Ring, airfoil body (Dwg 122-3-74).

Categories	Defects
Critical:	None defined
Major	AQL 1.0 percent defective
101	Leading edge not free from mold flash
102	Flash on other surfaces exceeds specified thickness and height
103	Durometer incorrect
104	Airfoil profile incorrect
105	Stations #2 or #13 diameters incorrect
106	Workmanship (3.5)

(c) Packaging (Section 5.2).

Categories	Defects
Critical:	None defined
Major	AQL 1.0 percent defective
101	Assembled improperly
102	Components missing or incorrect
103	Marking missing or incorrect

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(d) Packing (Section 5.3).

Categories	Defects
Critical:	None defined
Major	AQL 2.5 percent defective
101	Marking missing or incorrect
102	Component loose or missing
103	Shipping container incorrect

4.4.3.3 Packaging inspection. During or immediately prior to the packaging, 100 percent examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of another model blank cartridge or a dummy cartridge shall be classed as a major defect. Occurrence of any type other than those listed shall be classified as a critical defect. All nonconforming cartridges shall be rejected.

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Wrap assembly.

(1) Precondition assembled projectile for 24 hours at 130° ± 5°F.
(2) Cool projectile to room temperature in a low humidity environment (i.e., dessicator).
(3) Cut breakband and remove from projectile.
(4) Remove strip 8 wraps wide from breakband which was over cavity area of projectile. Breakband shall be tested within 5 minutes of removal from the projectile.
(5) Tape ends of test strip allowing 6 inches exposed.
(6) Mount test strip in tensile tester with jaws set 6 inches apart.
(7) Crosshead rate for test strip: 0.2 inches per minute.
(8) Recording chart rate: 10 inches per minute.
(9) Full scale of the chart shall initially be 20 pounds force. If the sample exceeds 20 pounds force, the scale shall be changed to 50 pounds force full scale without stopping the test.
(10) Calculations:

(a) Yield force, lbs - Maximum load reading on chart
(b) Yield strain, in/in - Draw a straight line that is best fit in the region between 5 and 10 pounds, and extend it to the zero load axis. The product of the distance along the zero load axis (y) from this point to the maximum yield point times the crosshead rate of the test strip divided by the product of the len: of the test strip times the recording chart rates equals yield strain.

Example: $\frac{Y \times 0.2}{6 \times 10} = \frac{Y}{300}$ yield strain

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4.4.4.2 Integrity/accuracy.

(a) Precondition assembled projectile for 24 hours at 130° ± 5°F.

(b) Cool projectile to room temperature in a low humidity environment (i.e., desiccator).

(c) The projectiles shall be fired from a M234 launcher assembled to a M16A1/AR15 rifle using M755 blanks in accordance with TM 9-1010-224-10. The rifle shall be secured in a fixed mount. Firing shall be conducted under nil wind conditions,

(d) The M234 launcher shall be cleaned and lubricated in accordance with TM 9-1010-224-10 prior to starting the test. A single "warming" round shall be fired prior to firing for record. Firing shall be in groups of 6 at the rate of one round every thirty seconds for the 6 rounds followed by a three-minute cool down period; repeat this cycle until all rounds are fired.

(e) Velocities shall be measured for information using photo electric ballistic screens and elapsed time counters. The distance from the center of the screens to the launcher barrel shall be 5.0 + 0.1 feet. If the velocity is above 200 or below 170 feet per second the shot shall not count.

(f) There shall be no breakband failures at launch as indicated by separation of the breakband from the projectiles prior to target impact and the projectiles shall impact within an area 46 cm wide by 175 cm high at the specified distance of 40 meters from the launcher. This shall be determined by the actual impact group size and dispersion and not by the aiming point.

(g) Removal and replacement of the launcher cup or use of a different laucher shall constitute a new test.

5. PACKAGING

5.1 Preservation.

5.1.1 Commercial/industrial. Projectiles shall be unit packed in a manner to prevent deformation or distortion by configuring the projectiles on a single layer, uniformly in a lay-flat configuration separated by interlocking partitions constructed from fiberboard material, sufficiently rigid to provide the support necessary to maintain the integrity of the projectile design. Additional requirements for the projectiles under this criteria shall be in accordance with ASTM D 3951.

5.1.2 Unit pack, level B. Six projectiles, together with six M755 blank cartridges (Dwg 122-3-43), shall be unit packed together within a carrier assembly as shown on Dwg 122-3-60.

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5.2 Packing, level B. Projectiles, unit packed as specified in 5.1.2, shall be packed together in multiples of four carrier assemblies, in accordance with Drawing 122-3-70.

5.3 Unitization. Unless otherwise specified in the contract or order, items packed in accordance with 5.2 shall be palletized in accordance with Drawings 19-48-4116-20PA1002 and 19-48-4116/32-20PA1002.

5.4 Marking.

5.4.1 Commercial/industrial. Projectiles packaged in accordance with 5.1.1 shall be marked in accordance with the applicable requirements of ASTM D 3951.

5.4.2 Level B. Projectiles, unit packed and packed in accordance with 5.1.2 and 5.2, shall in addition to any special marking required by the contract or purchase order be marked as appropriate in accordance with Drawings 122-3-143-10 and 122-3-70.

6. NOTES

6.1 Intended use. The projectile covered by this specification is intended for use in the M234 launcher.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

(c) Level of packaging.

6.3 Caution.

(a) The M755 Blank cartridges are classified class “C” explosives with a minimum of susceptibility to static discharge detonation. They should be handled in a careful manner with caution against impact and shock loading.

(b) Firing ranges shall be enclosed, constructed to contain the projectile and all access controlled during firing.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-51474A(EA)	PROJECTILE, 64 MM, RIOT CONTROL, KE M743

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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